

comes with the embedded 5.1 channels or by using 5.1-compatible soundcards. To satisfy these criteria we used the best 5.1-compatible soundcard, Creative Sound Blaster Live!

For the purpose of the test, we differentiated between the basic two-speaker setup (with or without external subwoofers) and surround speakers (4.1, 5.1 and higher speaker setups). It wouldn't make sense in pitching a lower end two-speaker setup with the higher surround speaker sets. You will find, on occasion, in the review that some four-speaker setups are significantly cheaper than some high-end two-speaker sets. In such cases it would always be preferable to go in for the four-speaker combinations (depending on your current soundcard and the room dynamics). What must be clearly understood is that, as with most things related to a PC, audio solutions too are interdependent. You may buy an excellent speaker set but if you miss out on the right soundcard then you will be compromising on the quality that your expensive speakers could have provided.

Two-speaker setups

Here we concentrated on speaker sets that come in the basic two-speaker models or the better 2.1 (two satellites and one woofer) models. This category covers the lower end of the market segment. The variety of sets available in this category is quite large and most of us end up getting these speaker sets along with our new PCs. Mercury had a strong presence in this segment with varied models. Typhoon has some very aggressively priced speaker sets in this category. A good development is the presence of Indian manufacturers such as Sumoko competing along with the heavyweights such as Creative, Samsung and Lexpro. The price spectrum ranges from Rs 375 to Rs 8,900.

Quality

In terms of quality of sound reproduced, **Typhoon Silver Crest 600** came out tops. The Typhoon Silver Crest gave decent audio quality while watching movies and offered excellent stereo effect. It also generated good quality treble and vocals. The quality of bass generated was average with slight distortion when we played music that had deep bass.

But the level at which the distortion occurs is tolerable as compared to the distortion generated by the Lexpro LS 880 or even the Typhoon DS 1032.

TEST PROCESS

The test system used to evaluate the speakers consisted of a Celeron 666-based computer with 128 MB SDRAM. The most important element of this configuration is the Sound Blaster Live! Platinum audio card used to test the speakers.

We tested the speakers using a carefully compiled collection of 36 audio tracks that cover the entire audio frequency range. The tracks used were made without any kind of electronic processing. They also contained subtle transients that define true hi-fidelity, which are sometimes lost in the reverberation and processing used in making conventional records. These tracks can not only test the performance of a wide range of sound equipment but also offer the very best in

musical, vocal and effects sources for experimental and demonstration purposes. The speakers were also examined by means of their performance during gaming sessions; we included the two most popular games—Quake III and Unreal Tournament.

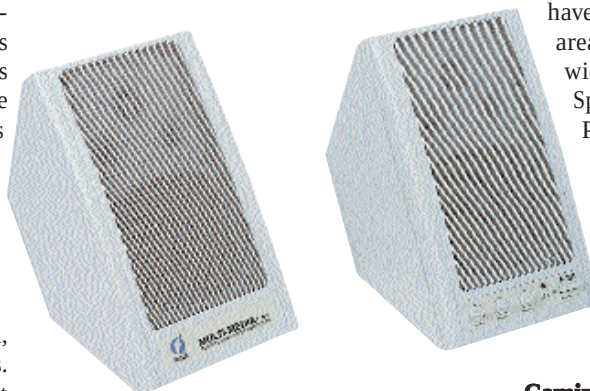
As speakers are also used while watching movies, to determine the quality of sound that the products would deliver, we used a series of clips that we judged could bring out the best or the worst in such a scenario. The clips included scenes from some popular movies. Again, care was taken that the test setup was arranged in a manner so as to prevent any of the characteristics of the room itself from interfering with the inherent characteristics of the speakers.

The runner up in this section was the Creative PC Works 2.1 that just marginally lost to the Typhoon SC 600 and was able to leave the Samsung SMS 8320 behind with a very comfortable margin. The Creative PC Works 2.1 did well in the treble and mid (vocals) frequency section and performed better with games and movies. However, while playing a sample that contained very low frequency, the reproduction was not satisfactory.

Unlike other stereo speakers wherein manufactures have a single full way

The Mercury SW480 was one of the cheapest and best-looking 2.1 speaker systems that we received. It is not high powered like the speaker systems mentioned earlier, but has the capability to deliver quality at those lower volumes.

The stereo section saw some entries that sported the NXT flat panel technology. NXT has developed a speaker system that comprises a flat panel. It has called this technology SurfaceSound. Unlike conventional speaker systems where the radiation of sound is very focussed (and you have a 'sweet spot') in your listening area, SurfaceSound provides a much wider audio listening angle. The Jazz Speaker Malibu and the Frontech Flat Panel, were two such speakers. They were able to reproduce good highs and mid frequencies but completely lacked the ability to produce bass. The bass actually sounded as if someone was trying to drum up a cardboard box.



Funai 788: Big bodied, with decent bass and mid-frequency response

speaker, the Samsung SMS 8320 has a separate tweeter and a woofer that helps in delivering more clarity in sound. Innovatively designed, this speaker set was able to produce the right amount of treble and mid frequencies, but its performance in the bass section was nothing to write home about.

Gaming

In the gaming section again, the **Typhoon Silver Crest 600** took away the laurels. There is ample power in this speaker system to suffice the needs of a high-end gamer. The speakers are capable of providing the perfect ambience required for playing games, with all those high pitched shrieks that your opponents make when you frag them in *Quake III*. Even the quality of bass is exceptionally good.